other radioactive methods, cf. Ch. 13), uranium content of crystals of several minerals in the rock, their fission track count and track-length distribution caused by thermal annealing. Zircon, titanite and apatite are examples of three such minerals in order of increasing annealing temperature. The temperature history of a rock formation is important in oil prospecting operations as the maximum temperature seems to be a useful indicator on whether to expect oil, natural gas or nothing. For too low temperatures neither oil nor gas is expected, for intermediate temperatures oil may be present and for high temperatures only natural gas.

9.2. GENERAL PROPERTIES OF DETECTORS

The principle for detection of nuclear radiation is the same as discussed previously for both the interaction of radiation with matter and radiation production namely: excitation, ionisation, deflection, and atom recoil. In the case of detection only the first two of the interactions are primarily used even if, naturally, the others also occur. A nuclear particle entering a detector produces excitation and ionization, both of which can be used for detection. When the excitation is followed by fluorescent de-excitation (section 7.8.4 and Ch. 8) the light emitted can be registered by light-sensitive devices, e.g. the photo-multiplier tube (PMT) which transforms the light into an electric current. Scintillation and Čerenkov detectors are based on light emission. A similar current is generated when production of *charge carriers* (i.e. ions, electrons and holes) takes place between the charged electrodes of a detector. Detectors based on the production of charge carriers are either gas-filled (ion chambers, proportional and Geiger-Müller tubes in which charge carriers are produced by ionization of a gas) or solid, usually semiconductor crystals. In the latter case electrons and holes are produced in pairs (section 9.4).

An ionizing particle or photon will produce a collectable charge ΔQ in the detector

$$\Delta Q = 1.60 \times 10^{-19} E_{\text{loss}} \eta w^{-1} \tag{9.2}$$

where E_{loss} is the total energy lost by the particle to the detector, η is the charge collection efficiency, w is the energy required for the formation of a pair of charge carriers in the detector medium, while the constant is the charge (Coulomb) of a single charge carrier (the pairs must be regarded as singles because they move in opposite direction in the electric field due to their opposite charges).

Because of the short duration of the absorption process for a single particle (10^{-4}—10^{-9} s) the short current, i, is referred to as a *pulse* of charge ΔQ

$$i = \Delta Q / \Delta t \tag{9.3}$$

If this current passes through a resistor R it will produce a voltage pulse (cf. Eqn (9.2))

$$\Delta V = R \, \Delta Q / \Delta t \tag{9.4}$$

The pulse is usually referred to as the *signal* (for the preamplifier). While the signal from semiconductor detectors is used as a charge pulse, most other detectors immediately convert the current to a voltage drop over a resistor. In either case, the output from the preamplifier is usually a voltage pulse. Preamplifiers are often integrated with the detector.

Pulse counting *per se* does not distinguish between different nuclear particles (α, β, γ, etc.) or between particles of different energy. Such distinction is obtained by choosing detectors of unique (or exceptionally high) sensitivity to the particles of interest. Energy analysis, if desired, is achieved by the accompanying electronic circuitry because the pulse charge or voltage (ΔQ or ΔV) is proportional to the energy of the absorbed particle.

9.2.1. Pulse generation

Figure 9.3 will be used to describe the formation of a voltage pulse. A detector is connected between points A and B. The detector has an *internal* resistance (because of the limited charge carrier mobility in the detector) and capacitance (because of mechanical construction), indicated by R_i and C_i. Figure 9.3(a) does *not* show the physical design of the detector, but only its electrical equivalents; this will make it easier to understand its function. When a particle enters the detector it produces charge carriers (this is symbolized by the closing of switch S), and the collection of these at the electrodes gives a current which, together with a small current from the bias supply through R_e, flows through R_i to ground. R_e is the resistance between the detector anode and the positive terminal of the bias voltage supply (voltage $+ V_0$); the other terminal is grounded. We shall concentrate our interest on the potential V_p at point P which is connected via the comparatively large capacitance C_e to the output. In general $R_e > R_i$ (under conducting

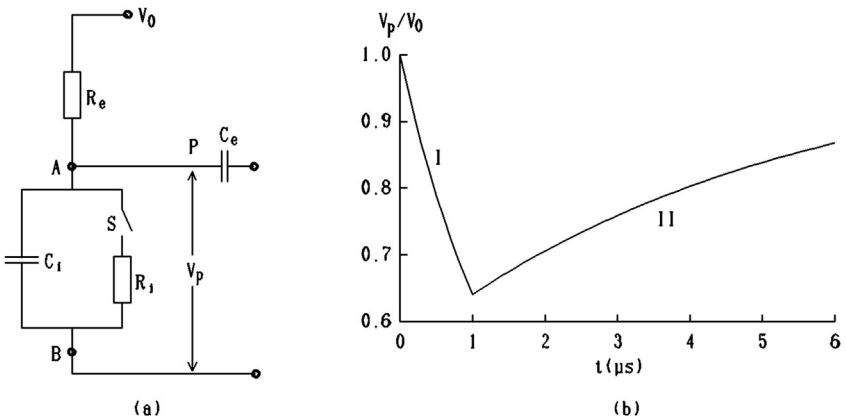


Figure 9.3 (a) Equivalent circuit (b) pulse shape in point P.

conditions), and $C_e \gg C_i$; for illustrative purposes we will assume $R_e = 50\ \text{k}\Omega$, $R_i = 10\ \text{k}\Omega$ and $C_e + C_i = 100$ pF.

When S is open (no ionization in the detector), the potential at point P must be $V_p = V_0$, i.e. the potential of the bias voltage. At time $t = 0$, S is closed (production of charge carriers has occurred in the detector because of interaction with a nuclear particle), and the charge of $C_i + C_e$ flows through R_i together with a small current from the bias supply through R_e. The potential in P decreases according to

$$V_p/V_0 = a\left(1 - e^{-bt}\right) + e^{-bt} \tag{9.5}$$

where $a = R_i/(R_i + R_e)$, $b = (a\,R_e\,C)^{-1}$, and $C = C_i + C_e \approx C_e$. For $R_e = 50\ \text{k}\Omega$, $R_i = 10\ \text{k}\Omega$ and $C = 100$ pF, $a = 1/6$ and $b = 12\,000\ \text{s}^{-1}$. $1/b$ is referred to as the *time constant* of the system (the decay time of the charge of the system); in the time $1/b$ the voltage has dropped to $V_0/\mathbf{e}$. Assuming the charge collection ends after 1 μs this is equivalent to opening switch S at that time. The voltage in P has now fallen to its minimum value as indicated in Figure 9.3(b).

When S is opened charge starts building up on the capacitors by current flowing from the bias voltage through resistor R_e. The voltage in P begins to increase again according to

$$V_p/V_0 = 1 - (1 - V_m/V_0)e^{-t/(R_e C)} \tag{9.6}$$

where V_m is the voltage in P when the switch is opened again, t is now the time after opening of S and $C = C_e + C_i$. If we assume $R_e = 50\ \text{k}\Omega$ the voltage build-up will follow the climbing curve to the right of 1 μs in Figure 9.3(b). In practice, the current flow stops more gradually and the potential in P follows a smoother curve.

The voltage drop at P results in a negative voltage pulse after the capacitor C_e. In Figure 9.3(b) the voltage change at point P is more than 40% of the bias voltage; this is highly exaggerated, since the change is normally a very small fraction of the bias voltage. This is due to the fact that the current is carried only by the charge carriers formed in the detector. If the number of primary charge carriers formed is n_i, the charge transport ΔQ is

$$\Delta Q = -1.60 \times 10^{-19} n_i a\ (\text{Coulomb}) \tag{9.7}$$

where a is the multiplication factor (1 in solid state detectors, $\gg 1$ in gas-filled detectors). When $a < 1$ it has the same meaning as the collection efficiency η in (9.2). The maximum voltage drop is then

$$\Delta V \approx \Delta Q/C \tag{9.8}$$

If 10^5 charge carrier pairs have formed in the detector and $C = 100$ pF, then $\Delta V = 0.16$ mV. In practice the effective voltage drop ΔV_{eff} is somewhat less than the calculated ΔV.

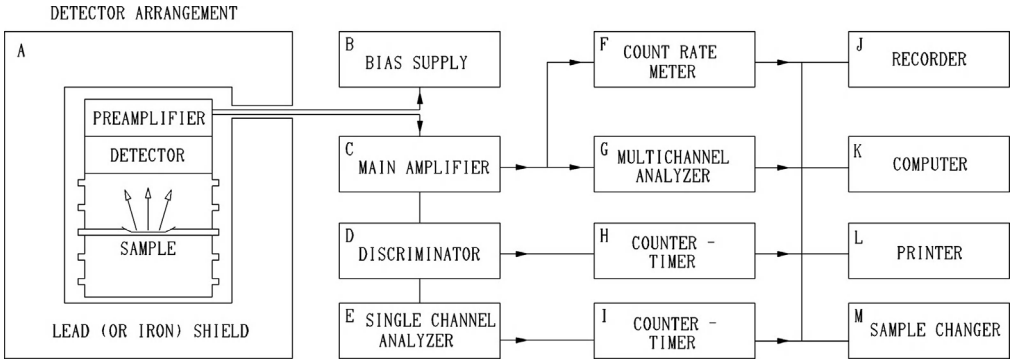


Figure 9.4 The most common pulse-type measuring units and combinations.

9.2.2. Basic counting systems

The block diagram of Figure 9.4 indicates the most common components in a measuring circuit. The detector (in its shield A) is connected to a stabilized bias voltage power supply B which furnishes the potential difference necessary for the detector to operate. The magnitude and type of the signal from the detector varies, depending on the type of detector, for the type of circuit illustrated in Figure 9.3 from a few hundred microvolts to several volts. In most cases it is necessary or preferable to have a *preamplifier* connected directly to the detector, cf. section 9.8.1. The preamplifier often increases the pulse size to the 0.1—10 V preferred for the auxiliary system. This initial amplifier also supplies the power needed to drive the load presented by the connecting cable and input impedance of the *main amplifier* C. The total gain in the system may vary from 10 for a Geiger-Müller counter to 10^4 for some solid state detectors.

A limit is set on the voltage gain of an amplifier by the presence of electronic noise at the amplifier input due to thermal noise in the detector and preamplifier input stage, microphonics, and components which cause small, random voltage changes. A part of this noise can be reduced by cooling the input circuitry. The noise is amplified with the signal and can mask small pulses from the detector. Some of the noise may be eliminated by proper design and operation of the electronic circuitry, but a small inherent noise is always present (Fig. 9.5(a) and section 9.8.1). The *signal to noise ratio* is a ratio of output pulse (for a given input signal) to the noise level at the output. Noise pulses can be rejected by a *discriminator* D which serves as a filter to allow only pulses of a certain minimum size to be passed on to the rest of the system; in Figure 9.5(a) the discriminator rejects all pulses below the dashed line. The signal pulse, after amplification, is sufficiently large to operate the electronic *counter* H in Figure 9.4 causing registration of the pulses. In some cases the number of pulses counted are often shown on light-emitting diode (LED), gas-plasma, or liquid crystal displays.

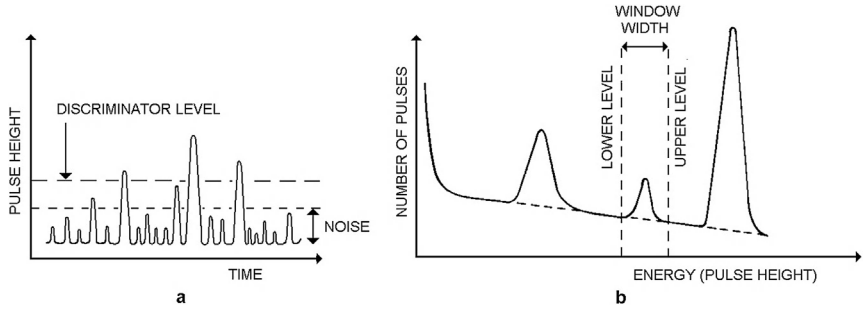


Figure 9.5 Idealized pulse spectrum obtained (a) from a shaping amplifier, and (b) from a multi-channel analyzer using a solid state detector. The figure shows the effect of different discriminator settings.

9.2.3. Pulse shape and dead time

From Figure 9.3(b) it is obvious that it takes some time to restore proper measuring conditions. Such conditions may be assumed to have occurred when the potential V_p has returned above a certain level. During the first part of this time the detector is unable to collect a new charge because working conditions have not been sufficiently restored. This time, which in the Figure is something like 10 μs, any new events would not produce a pulse crossing the discriminator level. This interval is properly called *dead time*, see Figure 9.6. Somewhat later the initial operating conditions are still not

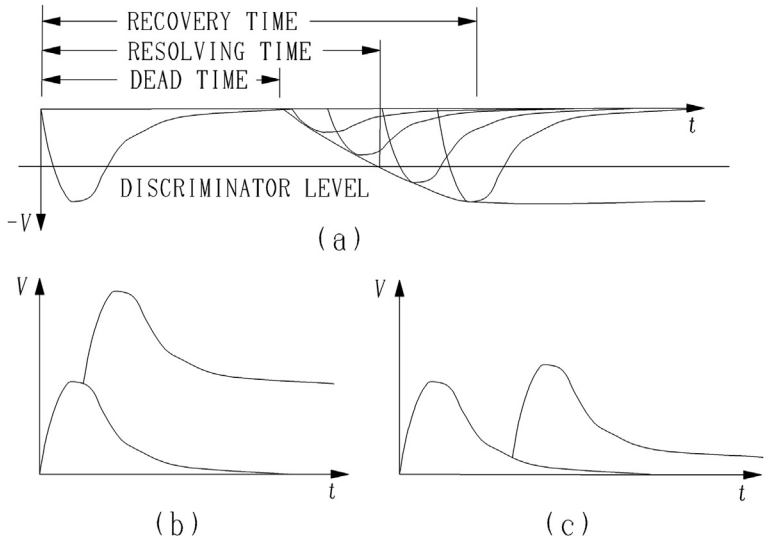


Figure 9.6 Effects of short interval between events registered in detectors. (a) G.M. tube, (b) peak pile-up and (c) tail pile-up for fast detectors.

fully restored but the detector is now able to produce a pulse of larger magnitude which triggers the discriminator. Still later, after the *recovery time*, the initial conditions are restored. If the second event occur within a short time, *peak pile-up* will occur. At a somewhat later time the new pulse overlaps with the tail of the pulse from the previous event causing so called *tail pile-up*. Pulse pile-up may make two or more closely spaced events look like a single more energetic event, Figure 9.6. The time needed to separate two events is referred to as *resolving time* (for simplicity dead time and resolving time will be used as synonyms in this text). Thus the detector and measuring circuit needs a certain time to register each individual event separately with correct magnitude. In many cases the measuring circuitry is much faster than the detector and the dead time is a function of the detector only. Since radioactive decay is a statistical random process and not one evenly spaced in time, see section 9.11, even for relatively low count rates a certain percentage of events will occur within the resolving time of the system. In order to obtain the true count rate it is necessary to know the correction that must be made for this *random coincidence loss*. In systems using a MCA for pulse height analysis, the MCAs pulse conversion time is usually determining the system dead time and not the detector.

Two different models exist for the dead time, t_d, of counting systems depending on system behavior after a pulse, see Figure 9.7. In a *nonparalyzable* system the pulses following the first within the dead time are lost, but the system is ready to accept another event immediately after the dead time has expired. If N_{obs} is the counted number of pulses during the measurement time, t_{meas}, and R_{real} the real number of particles which hit the detector per unit time, then the total lost time, t_{lost}, is

$$t_{lost} = N_{obs} \times t_d \tag{9.9}$$

and

$$N_{obs} = R_{real} \times (t_{meas} - N_{obs} \times t_d) \tag{9.10}$$

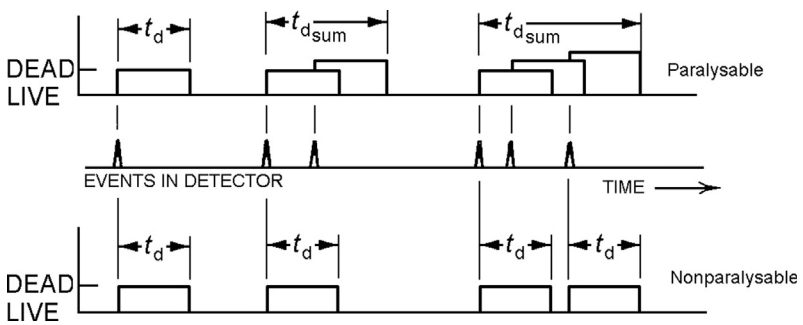


Figure 9.7 Paralysable and nonparalysable detector systems

but

$$R_{obs} = N_{obs}/t_{meas=} = R_{real} \times (1 - R_{obs} \times t_d) \tag{9.11}$$

thus

$$R_{real} = R_{obs}/(1 - R_{obs} \times t_d) \tag{9.12}$$

In a *paralyzable* system each event starts a new dead time period whether or not it generates an output signal. The dead time for the paralysable detector is a bit more tricky to estimate. However, assume that $P_1(t) \times dt$ is the probability that the next event will happen during the short interval dt after a delay of t,. Further more, assume that $P(0)$ is the probability that no event will occur between 0 and t and $R_{real} \times dt$ is the probability of an event occurring during dt. Then

$$P_1(t) \times dt = P(0) \times R_{real} \times dt \tag{9.13}$$

Poisson statistics gives that

$$P(0) = e^{-R_{real} \times t} \tag{9.14}$$

and hence

$$P_1(t) \times dt = R_{real} \times e^{-R_{real} \times t} \times dt \tag{9.15}$$

Assume now that R_{obs} is the count rate of time intervals betwen real events spaced more than t_d apart. The probability for intervals longer than t_d, $P_2(\Delta t > t_d)$, are given by integration of:

$$P_2(\Delta t > t_d) = \int_{t_d}^{\infty} P_1(r) \times dt = e^{-R_{real} \times t_d} \tag{9.16}$$

The rate of such events is then given by multiplying with R_{real} yielding:

$$R_{obs} = R_{real}\, e^{-R_{real} t_d} \tag{9.17}$$

At very low count rates it can be shown the result is independent of the type of system, i.e. $R_{obs} \approx R_{real}\ (1 - R_{real}\ t_d)$. However, the behavior of these two system types at high count rates are different. A nonparalyzable system shows an asymptotic approach to a maximum count rate with increasing source strength whereas the count rate on a paralyzable system passes through a maximum and then decreases again due to the exponential term in Eqn (9.17). Hence each reading on a paralyzable system corresponds to one of two values, one low and one high. Dangerous mistakes can occur by misinterpreting the reading from a paralyzable dose rate meter.

Most pulse height analyzer systems (MCAs) contain a dead-time metering circuitry. This is especially needed for MCA:s with varying dead-time (see section 9.8.5).

The simplest technique for measuring the resolving time t_r of a nonparalyzable counting system uses a method of matched samples. Two samples of similar counting rates are counted separately and then together. The combined sources should give about 20% fractional dead time, $R_{obs}\, t_r$. From the difference between the measured count rate of the pair together and the sum of their individual rates, the resolving time is calculated using the equations:

$$x = R_a\, R_b - R_{ab}\, R_0 \tag{9.18}$$

$$y = R_a\, R_b\, (R_{ab} + R_0) - R_{ab}\, R_0\, (R_a + R_b) \tag{9.19}$$

$$z = y\, (R_a + R_b - R_{ab} - R_0)/x^2 \tag{9.20}$$

$$t_r = x\left[1 - (1 - z)^{1/2}\right]/y \tag{9.21}$$

where R_a, R_b, and R_{ab}, are the measured count rate of samples a and b separately, a and b together, respectively. R_0 is the background count rate for the system. The correction for the dead-time can then be made according to Eqn (9.12).

A more accurate technique is based on the use of a short-lived radionuclide, e.g. ^{99m}Tc or ^{116m}In. The count rate is then measured a number of times during at least one half-life with the source left untouched in position all the time. When the background count rate can be neglected (which is usually the case) combining Eqn (9.12) with the equation for radioactive decay gives after some algebra

$$R_{obs}\, e^{\lambda t} = R_0 - R_0\, t_d\, R_{obs} \tag{9.22}$$

where t is the time of measurement, λ the decay constant and R_0 the initial count rate. A plot of $R_{obs}\, e^{\lambda t}$ as function of R_{obs} should yield a straight line with slope $-R_0\, t_d$ and intercept R_0 on the vertical axis. The dead time is then obtained as the absolute value of slope divided by intercept. If the plotted data deviates strongly from a straight line it indicates that the system investigated is paralyzable.

A peculiar property of nuclear decay is that events do not occur evenly distributed in time but as bursts of various amount of particles within short time intervals. The reason for this is found in the independence of each individual decay and is a side effect of the law of radioactive decay as derived in section 5.11. Accordingly the most probable time interval between any two decays is in fact zero and independent of the actual average decay rate! This leads to the requirement that nuclear pulse handling equipment must be able to separate events very close in time, i.e. have an extremely high band width.

9.3. GAS COUNTERS

All gas-filled counters are in principle ion chambers (with the exception of the less common gas scintillation counters). The ionization produced in an ion chamber by a single nuclear particle produces too low a charge pulse to be easily detectable except for α- particles. However, an ion chamber can be designed so that the number of ion pairs formed in each event is multiplied greatly.

Consider an ion chamber with a hollow cylindrical cathode and a thin central wire as an anode (Fig. 9.9(a) shows an old type GM-tube). The ion pairs formed in the gas by the passage of the ionizing radiation are separated from each other by their attraction to the electrodes. The small, very mobile electrons are rapidly collected on the anode, which is maintained at a high positive potential above ground, $\geq$1000 V. Most of the voltage pulse which appears on the anode arises by induction by the positive ions as they move away from the immediate region of the anode. This step is responsible for the rise time of the pulse, curve I in Figure 9.3(b). The potential decrease is only momentary as the anode is rapidly recharged by the power supply. The time necessary to restore the original potential is a function of the decrease in electric field near the anode due to the build-up of a layer of slow-moving positive ions, and of the time constant circuitry (curve II in Fig. 9.3(b)).

The electric field strength at a distance x from the anode is proportional to $1/x$. If the applied voltage of a cylindrical chamber of 1 cm radius is 1000 V, the potential in the immediate vicinity of a center wire of 0.0025 cm diameter is approximately 7×10^4 V cm^{-1}. As the primary electrons reach the vicinity of this high field and increase their kinetic energy, they cause secondary ionization which increases the pulse detected at the wire anode. The collected charge is given by Eqn (9.7), where a, *the gas multiplication factor*, is $\gg 1$.

This gas multiplication factor varies with the applied voltage as illustrated in Figure 9.8. In region II of the curve a flat plateau over a relatively wide voltage range is observed. Prior to the attainment of the threshold voltage for the plateau, the ions would not have sufficient *drift velocity* to prevent elimination of some ion pairs by recombination. Throughout the plateau range the drift velocity is sufficient to make recombination negligible and since secondary ionization is not present, $a = 1$. Ionization chambers operate in this voltage region. In region III the electrons from the primary ion pairs receive sufficient acceleration to produce additional ionization and the process of gas multiplication ($a > 1$) increases the number of collected charges. This is the region of *proportional counter operation* as the pulses are proportional in size to the energy deposited in the detector by the passage of the initial radiation. Region IV is the one used for *Geiger-Müller counter* (or GM counter) operation. In this region the gas multiplication is very high ($\geq 10^6$) and the pulse size is completely independent of the initial ionization. Beyond region IV continuous discharge in the detector occurs.